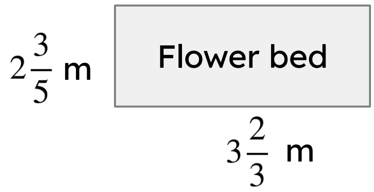
Problem solving with arithmetic procedures involving fractions

Task A: Solving problems involving fractions

1a) A rectangular flower bed has length $3\frac{2}{3}$ m and width $2\frac{3}{5}$ m. Jacob wants to plant flowers in the plot, but needs to buy the compost. The compost bags each cover 1 m^2 and cost £5.70 per bag. How much will it cost Jacob?

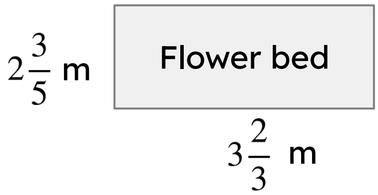


$$3\frac{2}{3} \times 2\frac{3}{5}$$

$$\frac{11}{3} \times \frac{13}{5} = \frac{143}{15} = 9\frac{8}{15}$$

Jacob needs to buy 10 bags each costing £5.70 so it will cost Jacob £57

b) Jacob also needs to put a metal edging around the plot. The edging is sold in 1 m lengths for £3.42 per metre. How much will it cost Jacob to put the edging around the flower bed?



$$3\frac{2}{3} + 2\frac{3}{5} + 3\frac{2}{3} + 2\frac{3}{5} =$$

$$3 + 2 + 3 + 2 = 10$$

$$\frac{2}{3} + \frac{3}{5} + \frac{2}{3} + \frac{3}{5} =$$

$$\frac{10}{15} + \frac{9}{15} + \frac{10}{15} + \frac{9}{15} = \frac{38}{15} = 2\frac{8}{15}$$

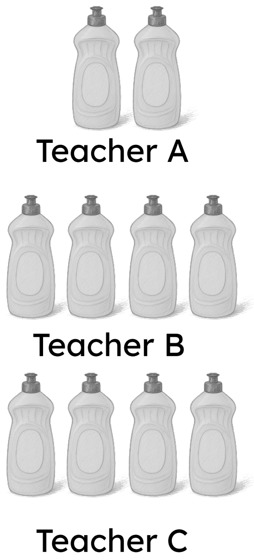
$$10 + 2\frac{8}{15} = 12\frac{8}{15}$$

$$13 \times 3.42 = 10 \times 3.42 + 2 \times 3.42 + 1 \times 3.42$$

$$34.20 + 6.84 + 3.42 = 44.46$$

Jacob needs to buy 13 m of fencing at £3.42 a metre. The total is £44.46

For the following question, do not use a calculator and show all your working out.



2) A bottle contains $\frac{3}{4}$ litre of water. On a school trip, three teachers have a different number of water bottles to give to the students in their groups. Each student gets an equal share.

a) How much water does each teacher have to share?

Teacher A $\frac{3}{2}$ *litre*

Teacher B and C each have 3 litres.

How much water do the students in each group get if:

b) There are three students in Teacher A's group?

$$\frac{3}{2} \div 3 = \frac{1}{2}$$

Each student gets $\frac{1}{2}$ *litre each.*

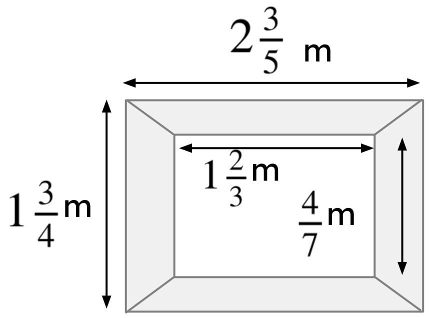
c) There are six students in Teacher B's group?

Each student gets $\frac{1}{2}$ *litre each.*

d) There are five students in Teacher C's group?

Each student gets $\frac{3}{5}$ *litre each.*

Task B: Using a calculator



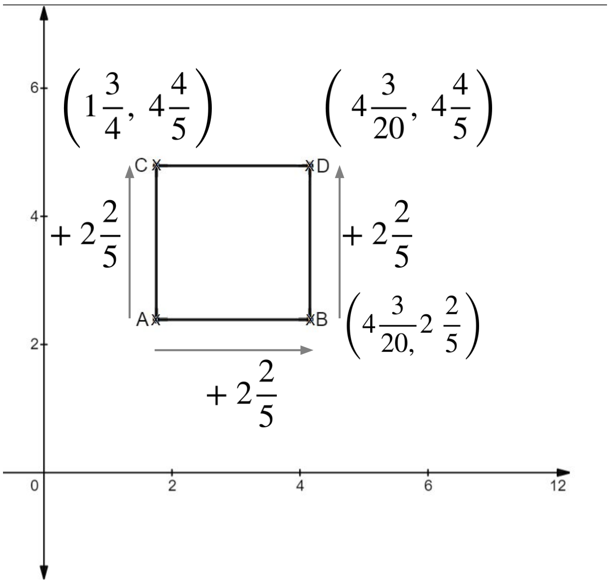
1) Aisha is restoring some old picture frames. She wants to cover the frame in gold leaf.

a) Work out the area of the frame.

b) Gold leaf costs £12.84 per m^2. How much will it cost Aisha to gold leaf the picture frame?

$$\left(2\tfrac{3}{5} \times 1\tfrac{3}{4}\right) - \left(1\tfrac{2}{3} \times \tfrac{4}{7}\right) = \frac{1511}{420} = 3\frac{251}{420}$$

Therefore, 4 m^2 *sheets are needed. The total cost is 4 × £12.84 = £51.36*



2) A square with length $2\frac{2}{5}$ units has been plotted on an axis.

The coordinate of A is $\left(1\frac{3}{4}, 2\frac{2}{5}\right)$

Write the coordinate of B, C and D, giving your answer as a mixed number.

B $\left(4\frac{3}{20}, 2\frac{2}{5}\right)$

C $\left(1\frac{3}{4}, 4\frac{4}{5}\right)$

D $\left(4\frac{3}{20}, 4\frac{4}{5}\right)$